Chapter 1 : Introduction to DBMS

Q1. What is DBMS ?

A. DBMS is an environment that is both convenient and efficient to use. It consists of a collection of interrelated data & a set of programs to access the data. Databases are used to manage large amounts of data efficiently and allow users to retrieve, update, and manipulate data in various ways.DBMS provides an interface between the user and the database, allowing users to interact with the data stored in the database.

It has applications in :
- Banking
- Airlines
- Universities
- Sales
- Manufacturing
- HR

Q2. State the Properties and characteristics of DBMS ?

A. Some characteristics of DBMS :
- Data security
- Removes redundancy
- Self-describing nature
- Multiple views of data
- Data sharing
- Multi user transaction processing

DBMS has ACID Properties :
A - Atomicity
C - Consistency
I - Isolation
D - Durability

Q3. State what are the Drawbacks of the file system ? **OR**
State the Advantages of DBMS ? **OR**
Differentiate between File System and DBMS.

A.

1. Data redundancy(Duplication) and inconsistency
   Multiple file formats, duplication of information in different files.
2. Difficulty in accessing data
   Need to write a new program to carry out each new task.
3. Data isolation
   Multiple files and formats.
4. Integrity(Correctness) problems
   - Integrity constraints (e.g., account balance > 0) become "buried" in program code rather than being stated explicitly.
   - Hard to add new constraints or change existing ones.
5. Atomicity of updates
   Failures may leave database in an inconsistent state with partial updates carried out.

6. Concurrent(Multiple at the same time) access by multiple users
   Concurrent access needed for performance as uncontrolled concurrent accesses can lead to inconsistencies.

7. Security problems
   Hard to provide user access to some, but not all, data.

A. The DBMS system architecture consists of 4 major parts :

1. Disk storage
2. Storage Manager
3. Query processor
4. End user

1. Disk Storage :
It consists of DIDS :
Data, Indices, Data dictionary, Statistical data.

2. Query Processor :